

제안발표 대본

p1.

Hello, I'm **Jin Kim**, the presenter for **Team 1**.

Today, we will present our project, **Urban Spatial Data-Based Personalized Running Route Recommendation System**.

Let's begin.

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The presentation will proceed in these following orders.

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Running has steadily grown in popularity over the past two years, and now Korea is experiencing a real running boom.

In just one year, running participation increased by about 60%, and more than 130 new marathon events were added.

Now, it is common to see people running in parks and around our neighborhoods.

So, running has become more than just exercise. It has become a lifestyle.

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However, as running becomes more popular, several pain points have also emerged.

Even when runners want to try a new route, they often end up repeating familiar routes or searching for routes themselves through social media, blogs, and maps.

Popular running spots can also become crowded, making the running experience less comfortable.

One of our team members, who is an actual runner, also mentioned that searching for running routes one by one while traveling was inconvenient, and that popular routes were sometimes too crowded.

So, what runners really need is not just a popular route, but **an easy way to find a new route that fits them right now**.

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So, is it enough to simply recommend a new route?

Not really.

Even for the same 5 kilometers, different runners have different goals, so the criteria for a good route also change.

For example, a runner who wants to maintain pace may prefer flat roads with fewer signals, while a runner who wants to build endurance may prefer uphill sections and higher elevation gain.

At night, lighting and pedestrian safety become more important, while some runners may prefer greener or more scenic routes.

So, there is no single best running route. **The best route depends on each runner's goal and preference.**

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As running becomes more popular, there are now many running apps.

However, we found that only a few apps actually recommend running routes, and it was even harder to find services that generate routes based on each runner's goal.

Most existing apps mainly focus on tracking distance, time, pace, or providing training features.

So, we saw a need for a service that focuses not only on **how much you run, but also on where you should run**.

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So, to get straight to the point, our goal is to build **DALRO, a personalized running route recommendation app**.

DALRO provides four main features.

First, it **automatically generates personalized routes** based on the starting point, target distance, and running goal.

Second, it provides **goal-based recommendations** such as PACE, POWER, NIGHT, and GREEN.

Third, it offers **multiple route options** that match the selected conditions.

Finally, it provides **route environment information**, such as distance, slope, signals, lighting, greenery, and crowding.

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Our target area is **Seoul**.

Seoul provides relatively easy access to spatial data such as road networks, living population, green spaces, waterfronts, and traffic volume.

It also has diverse running environments, including the Han River, parks, urban areas, and hills.

In addition, running demand is high, so we considered Seoul a suitable area for generating and validating different types of routes.

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We expect three main final outputs.

First, **road-level running suitability data** for Seoul.

Second, a **goal-based route generation algorithm** that creates routes based on the user's running purpose.

Third, the **DALRO application and validation results**, which turn these outputs into a real service.

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We will use several types of urban spatial data.

These include road networks, elevation and slope, signals and intersections, streetlights, living population, traffic volume, and green and waterfront areas.

Rather than choosing the data first, **we started from the types of routes we want to recommend and worked backward to identify the required data**.

For example, PACE focuses on flat roads and fewer signals, POWER on slope and elevation gain, NIGHT on lighting and pedestrian conditions, and GREEN on green and waterfront environments.

So, we selected the spatial data needed to evaluate these characteristics quantitatively.

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We divided the implementation process into **four steps**.

First, we collect and preprocess the spatial data.

Second, we analyze the running suitability of each road segment.

Then, we apply goal-based weights to develop the route recommendation algorithm.

Finally, we integrate everything into the DALRO app service.

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We plan to calculate running suitability in the following way.

First, we will build a runnable road network using OpenStreetMap.

Then, we will create buffers around each road segment and combine spatial data such as streetlights, signals, intersections, population, and green spaces using spatial joins.

We will also use DEM data to calculate terrain features such as slope and cumulative elevation gain.

Finally, we will normalize these indicators from zero to one and assign running attributes to each road segment.

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We plan to implement the goal-based route recommendation in the following way.

First, we assign different weights to road attributes depending on the running goal, such as PACE, POWER, NIGHT, or GREEN.

Then, we apply these weights to the road attributes and calculate an edge cost for each road segment.

Finally, we use Dijkstra 알고리즘 to search candidate routes, apply conditions such as starting point, target distance, and loop or out-and-back type, and recommend the top two or three routes that best match the user's goal.

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We plan to develop the app with this architecture.

For the frontend, we will use **React Native** to build a cross-platform app for both iOS and Android.

For the map, we will use the **Kakao Maps API and SDK** to visualize recommended routes.

For the database, we will use **PostgreSQL with PostGIS** to retrieve road and spatial data through spatial queries.

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This is Our expected schedule.

By the midterm presentation, we plan to build the **app MVP with one working recommendation mode**, and we are currently considering PACE as the first mode.

Then, by the final presentation, we will add the remaining POWER, NIGHT, and GREEN modes, complete the app, and aim to release it as a real service.

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We divided the roles as follows.

Kyungsoo Yoo is responsible for **data acquisition and preprocessing**, Siwon Kim for **running suitability analysis**, Yujin Kim for the **route recommendation algorithm**, and I am responsible for **application development**.

Yujin Kim is also our team leader and an actual runner, so she will conduct field tests by running the recommended routes herself.

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We plan to validate the system in three ways.

First, we will compare our recommended routes with a **baseline route** under the same starting point and target distance.

Second, we will quantitatively check whether the goal-specific metrics actually improve.

For example, PACE should have fewer signals and less slope variation, while POWER should have higher average slope and elevation gain.

Finally, actual runners will test the recommended routes and evaluate GPS accuracy, goal suitability, and running satisfaction.

So, we will validate our recommendations through **both quantitative comparison and real-world field testing**.

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We expect four main benefits.

First, DALRO can reduce the burden of manually searching for running routes.

Second, it can provide a more personalized running experience based on goals such as pace, endurance, night safety, or scenery.

Third, by suggesting alternative routes, it can help reduce crowding on popular running spots.

Finally, the service can be expanded from Seoul to other regions, and even to walking or cycling services.

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That's the end of our presentation. Thank you.

우리팀한테 들어올 질문

1. Will the spatial analysis and route generation run directly on the mobile app?

공간분석과 코스 생성 알고리즘을 모바일 앱에서 직접 실행할 예정인가요?

Answer — EN

No. Most spatial analysis will be processed in advance and stored in PostGIS. The mobile app will only send user inputs such as location, distance, and running goal to the backend. The server will run the recommendation algorithm and return the generated routes to the app. This keeps the app lightweight and reduces response time.

답변 — KR

아닙니다. 대부분의 공간분석은 미리 수행한 뒤 결과를 PostGIS에 저장할 예정입니다. 앱에서는 위치, 목표 거리, 러닝 목적 같은 사용자 조건만 서버로 보내고, 백엔드에서 추천 알고리즘을 실행해 생성된 코스를 앱으로 반환하는 구조입니다. 이렇게 하면 앱의 연산 부담을 줄이고 응답 속도도 확보할 수 있습니다.

2. What if there is no route that exactly matches the target distance?

사용자가 5km를 입력했는데 정확히 5km인 코스가 없으면 어떻게 하나요?

EN

We will allow a small distance tolerance and generate multiple candidate routes. Then, we will rank them based on both distance error and running suitability, rather than requiring an exact match.

KR

목표거리를 정확히 맞추는 것보다는 일정 허용 범위를 두고 여러 후보 코스를 생성할 예정입니다. 이후 목표거리 오차와 러닝 적합도를 함께 고려해 가장 좋은 후보를 추천합니다.

3. Is one runner enough to validate the system?

팀원 한 명이 직접 뛰어보는 것만으로 검증이 충분한가요?

Answer — EN

No. The field test is only a supplementary validation method. Our main evaluation will be based on quantitative comparisons with baseline routes. For the field test, we also plan to recruit additional runners, and if necessary, contact a running club at Inha University to obtain more user feedback.

답변 — KR

아닙니다. 실제 러너의 주행 테스트는 보조적인 검증 방법이고, 핵심 검증은 Baseline Route와의 정량적 비교입니다. 실주행 평가도 팀원 한 명으로 끝내지 않고 추가 러너를 모집할 예정이며, 필요하다면 인하대학교 러닝 동아리에도 협조를 요청해 더 많은 사용자 피드백을 확보할 계획입니다.

우리가 2조한테 할 질문

1. How will you account for changes in the urban environment over such a long study period?

한국어

연구 기간이 2010년부터 2025년까지로 장기간인데, 이 기간 동안 토지이용이나 하수관로 용량 등 서울의 도시환경이 많이 변화했을 것으로 보입니다. 머신러닝 학습 시 이러한 시간에 따른 환경 변화를 어떻게 반영할 계획인가요?

English

Your study covers a long period from 2010 to 2025, during which Seoul's urban environment, such as land use and drainage capacity, may have changed significantly. How do you plan to account for these temporal changes when training the machine learning model?

2. Why did you choose machine learning for this study?

한국어

침수 예측 방법 중 머신러닝 모델을 주로 채택하신 특별한 이유가 있나요?

English

Why did you choose machine learning as the main approach for flood prediction in this study?

조금 더 깊게 물으면:

What advantages does machine learning offer over conventional statistical or hydrological approaches for this study?

이 버전이 질문으로는 더 좋음.

3. How will you address the limitations of flood trace maps as training data?

한국어

침수흔적도는 실제 피해지역을 사후 조사해 작성되기 때문에 강우의 상세한 시공간 분포나 실제 침수심을 완전히 반영하지 못할 수 있습니다. 학습 데이터의 정확도를 높이기 위한 별도의 검증이나 보정 방법이 있나요?

English

Flood trace maps are generally produced after flood events based on surveyed damage areas, so they may not fully capture detailed rainfall patterns or actual flood depths. Do you have any additional validation or correction process to improve the reliability of the training data?